# Vinith Kuruppu

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## Education

**University of California, Berkeley** - Master of Science in Data Science (GPA: 3.96/4.0) Dec 2025

- **Relevant Coursework:** Applied Machine Learning, Natural Language Processing, Computer Vision, Generative AI, Research Design and Application, Data Engineering, Experiments and Causal Inference

**Loyola University Chicago** - Bachelor of Science in Computational Neuroscience May 2022

## Skills & Technologies

**Languages:** Python, R, SQL, Bash
**Machine Learning & AI:** PyTorch, TensorFlow, Transformers (Hugging Face), scikit-learn, XGBoost, QLoRA/LoRA/PEFT fine-tuning, RAG, Prompt Engineering, Vision Transformers, CNNs, ASR, audio preprocessing, FAISS, dense/sparse/hybrid retrieval, cross-encoder reranking
**Research Methods & Statistical Methods:** Ablation Studies, Benchmark Creation, Experimental Design, A/B testing, Causal Inference, Regression Modeling, Hypothesis Testing, Effect Sizes & Confidence Intervals, Power Analysis, Fairness Analysis & Bias Mitigation
**Data, Infrastructure & Visualization:** AWS (SageMaker, EC2, S3), Pandas, NumPy, Docker, Git, PostgreSQL, MLflow, Weights & Biases, MongoDB, Neo4j, Redis, Vector Databases, Matplotlib, Seaborn, Tableau

## Experience

**Graduate Student Instructor**, University of California, Berkeley Jan 2025 – Present

- Mentored 80+ master's students on research design and ML applications, improving average project scores by 12% through structured guidance on experimental design, causal inference, and statistical rigor
- Delivered feedback on 40+ research projects involving predictive modeling, A/B testing, and causal analysis, raising proposal quality by 20% as measured by rubric scores

**AI/ML Research Associate**, RAND Corporation May 2024 – May 2025

- Developed fairness-aware allocation models that directed $70 billion across 3,300+ hospitals, reducing variance between need/funding and improving equity in disaster reimbursement across US regions
- Engineered multi-class classification models achieving a 22% accuracy improvement over baseline through custom feature engineering and optimization, processing 1M+ expense records; the taxonomy and cost-ceiling estimates became the standard framework for FEMA allocation reviews
- Designed simulation frameworks to model disaster-response scenarios under uncertainty, conducting sensitivity analyses across 50+ parameter configurations; findings shaped federal preparedness policy recommendations for FEMA leadership
- Authored 30+ technical reports for congressional committees and federal agencies, translating complex statistical findings into policy recommendations that drove ongoing quality assurance improvements in multi-billion-dollar allocation processes

**Research Data Scientist**, Exponent Incorporated July 2022 – May 2024

- Built Python pipelines processing high-frequency signals from 3,000+ Apple Watch biosensors for Apple Health's hypertension detection algorithm, reducing integration time by 15%; analysis prioritized design changes for a $12B product line, shortening decision cycles by 20% and improving signal quality pass rate by 25% through systematic hypothesis testing and statistical process control
- Led end-to-end FDA-compliant validation for biosensor clinical trials, achieving 100% pass rate on internal QA audits through rigorous statistical validation and bias analysis; work enabled product launch for a health monitoring platform
- Conducted fairness-aware analysis of biosensor signals across demographic subgroups (age, gender, skin tone), identifying algorithm performance disparities that informed design changes and improved signal quality pass rate

by 25%, ensuring equitable accuracy

- Diagnosed and resolved defects in proprietary study software, raising automated test coverage and accelerating release cadence by 50%, eliminating critical bugs in FDA-regulated clinical trial systems

**Computer Vision Research Assistant**, Loyola University Chicago Jan 2020 – June 2022

- Led a 6-person research team developing bio-inspired CNN architectures with attention mechanisms to model human visual perception, establishing lab benchmarks for human behavioral data from EEG/eye-tracking studies.
- Improved model accuracy by 11% over prior lab baseline through systematic hyperparameter optimization, targeted data augmentation, and architecture refinements, achieving state-of-the-art performance on object recognition tasks
- Engineered Python data pipelines for image and behavioral datasets, reducing preprocessing time by 30% and cutting experiment turnaround through parallelization and automated quality checks
- Developed technical documentation and contributed to grant proposals supporting continued funding

## Projects

**SurgiRAG: Retrieval-Augmented Generation for Surgical Question Answering** 2025

*PyTorch, LLaMA-11B (QLoRA), BioBERT, BGE, FAISS, Hugging Face – GitHub*

- Developed domain-adaptive RAG system achieving a 40-point improvement in factual accuracy over GPT-4 baseline for surgical question-answering, using BioBERT dense retrieval, BGE cross-encoder reranking, and LoRA-fine-tuned LLaMA-11B
- Designed novel LLM-as-judge evaluation framework using GPT-4o-mini to assess faithfulness vs. fluency trade-offs not captured by standard metrics (ROUGE/BLEU); the framework revealed SurgiRAG's superior faithfulness (0.66) vs. baseline (0.25)
- Conducted a 12-variant ablation study across retrieval, reranking, and generation modules and curated a 32-item benchmark with a 1,527-chunk surgical corpus, showing semantic retrieval plus cross-encoder reranking improves faithfulness by 40% over baselines in high-stakes medical QA

**PathoVision: Hybrid Vision Transformer for Brain Tumor Classification** 2025

*PyTorch, DINOv2 (Vision Transformer), scikit-learn, OpenCV – GitHub*

- Engineered hybrid computer vision pipeline combining handcrafted features (Canny edge detection, Difference-of-Gaussian) with DINOv2 Vision Transformer embeddings for multiclass brain tumor MRI classification, achieving 96.7% test accuracy and outperforming end-to-end CNNs on limited labeled data
- Systematically benchmarked 40+ combinations of feature sets and classifiers, showing that simple logistic regression on DINOv2 embeddings matches or exceeds deep CNN performance while training in seconds, validating frozen self-supervised features as a low-compute option for resource-constrained clinical settings
- Developed a multi-stage preprocessing workflow (CLAHE, skull stripping, normalization) to standardize heterogeneous MRI inputs and mitigate orientation, contrast, and compression artifacts; demonstrating that DINOv2 embeddings encode anatomical features without supervision, providing strong inductive bias for medical imaging

**AgeVoicE: Adapting Whisper for Disfluent Speech** 2025

*PyTorch, Whisper, LoRA, Hugging Face, AWS – Manuscript in Preparation*

- Fine-tuned Whisper ASR models for older adult and dementia-affected speech using LoRA parameter-efficient adaptation, achieving a 22% WER improvement over baseline and improving ASR accessibility and fairness for underserved populations in voice assistants and on-device applications
- Built an end-to-end training pipeline for disfluent speech by developing a custom preprocessing workflow (background noise, disfluencies, non-standard pronunciations, audio normalization) and running systematic hyperparameter sweeps (learning rates, LoRA ranks, dropout) with experiment tracking and AWS distributed training, identifying regimes where loss decreased but WER increased and revealing overfitting to acoustic features vs. linguistic content
- Performed systematic failure analysis of base and fine-tuned Whisper models on older adult and dementia-affected speech, stratifying by acoustic conditions and speaker characteristics and using regression modeling and hypothesis testing to identify factors significantly associated with elevated WER in this population; findings guided targeted data augmentation to correct this bias.